

MicroJava – Grammar

Program = "program" ident { ConstDecl | VarDecl | ClassDecl }
"{" {MethodDecl} "}".

ConstDecl = "final" Type ident "=" (number | charConst) ";".

VarDecl = Type ident { "," ident } ";".

ClassDecl = "class" ident "{" { VarDecl } "}".

MethodDecl = (Type | "void") ident "(" [FormPars] ")"
{ VarDecl } Block.

FormPars = Type ident { "," Type ident }.

Type = ident ["[" "]"].

Block = "{" { Statement } "}".

Statement = Designator (Assignop Expr | ActPars | "++" | "--") ";"
| "if" "(" Condition ")" Statement ["else" Statement]
| "while" "(" Condition ")" Statement
| "break" ";"
| "return" [Expr] ";"
| "read" "(" Designator ")" ";"
| "print" "(" Expr ["," number] ")" ";"
| Block
| ";".

Assignop = "=" | "+=" | "-=" | "*=" | "/=" | "%=".

ActPars = "(" [Expr { "," Expr }] ")".

Condition = CondTerm { "||" CondTerm }.

CondTerm = CondFact { "&&" CondFact }.

CondFact = Expr Relop Expr.

Relop = "==" | "!=" | ">" | ">=" | "<" | "<=".

Expr = ["-"] Term { Addop Term }.

Term = Factor { Mulop Factor }.

Factor = Designator [ActPars]
| number
| charConst
| "new" ident ["[" Expr "]"]
| "(" Expr ")".

Designator = ident { "." ident | "[" ["~"] Expr "]" }.

Addop = "+" | "-".

Mulop = "*" | "/" | "%".